

Installing Quartus Prime, ModelSim, and FPGA Device Support

Notes for installing the software below:

- On Windows, after downloading an .exe installation file and clicking "Run", if a dialog pops up saying it was prevented from running, click "More Info" > "Run Anyway"
- On Mac, if you get a security message when you try to open a .dmg installation file, go to System Preferences > Security and Privacy > General, and under "Allow apps downloaded from:" click the button on the right to "open anyway"

NOTE: VirtualBox is not currently compatible with M1 (ARM-based) Macs – if you have a Mac with the M1 processor, it is recommended that you work on the lab computers or request to borrow a device from Pitt: <https://www.technology.pitt.edu/remotedevices>

Part I: (Mac Users Only) Install VirtualBox

1. Download VirtualBox installer from: <https://www.virtualbox.org/wiki/Downloads>
2. Click the download link for "OS X hosts" under "VirtualBox 6.1.12 platform packages"
3. Click the download link under "VirtualBox 6.1.12 Oracle VM VirtualBox Extension Pack" (needed for USB device support)
4. Open each of the downloaded files and follow the steps to install
5. Download a Windows 10 .iso to install on the virtual machine. You have two options:
 - a. via Pitt: <https://pitt.onthehub.com>
 - b. via Microsoft: <https://www.microsoft.com/en-us/software-download/windows10ISO>
6. Open VirtualBox and click "New" to create a Windows 10 virtual machine (if you name the machine "Windows 10" it will automatically select Windows 10 as the OS version)
7. Follow the steps to set up the virtual machine with default selections, **except** assign 100GB virtual hard disk storage instead of the default 50 GB.
8. Start the virtual machine (click the name of the virtual machine and then the arrow labeled "Start") and select the location of the Windows 10 .iso that you previously downloaded
9. Follow the steps to install Windows on the virtual machine
10. Complete all remaining software installation steps below on the Windows virtual machine
11. It is recommended that you periodically back up files you create on the virtual machine, e.g., by uploading files to OneDrive

Part II: Install Quartus Prime 18 Lite, ModelSim, and MAX 10 FPGA Device Support

Note for Mac users: Perform all steps below on a Windows virtual machine.

1. Go to <https://www.intel.com/content/www/us/en/software-kit/665990/intel-quartus-prime-lite-edition-design-software-version-18-1-for-windows.html> to download Quartus Prime 18 Lite. Note that newer editions should also work, but 18 is least likely to have unexpected issues.
2. Under "Individual Files", click the download buttons for ModelSim-Intel FPGA Edition and Intel Quartus Prime, and under "Devices" click the download button for Intel MAX 10 Device Support. If a legal disclaimer appears, click "Agree" – this is related to this being an older version of the software. The files are large so the download may take some time. When you first try to download, you will be

asked to create an account - register for an account. You should receive a confirmation email, but it may take some time. You can try logging in and restarting the download in the meantime. NOTE: If the download still does not start, you may need to try a different browser, e.g., Internet Explorer.

Downloads

[<](#) [Multiple Download](#) [Individual Files](#) [Additional Software](#)

Intel® Quartus® Software

ModelSim-Intel® FPGA Edition (includes Starter Edition)

Download
ModelSimSetup-18.1.0.625-windows.exe

Size: 1.1 GB
SHA1: f4b428584c780016d119c0b1fd16c26dee880dcc

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Intel® Quartus® Prime (includes Nios® II EDS)

Download
QuartusLiteSetup-18.1.0.625-windows.exe

Size: 1.7 GB
SHA1: 70faf36e2c8d69aa5243de767242a75832fa749e

▼

Under “Devices”:

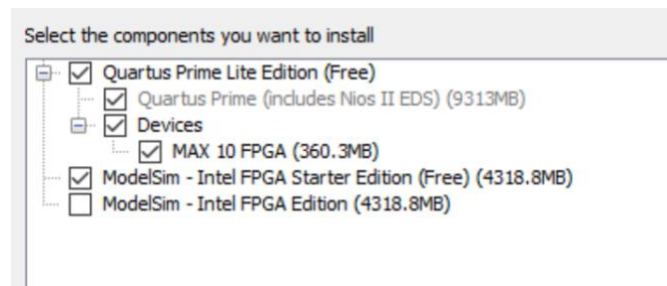
Intel® MAX® 10 Device Support

Download
max10-18.1.0.625.qdz

Size: 330.9 MB

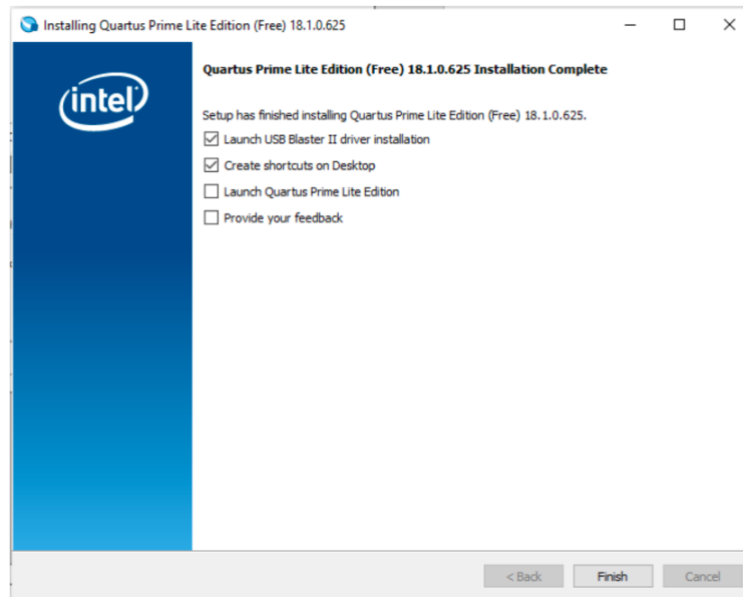
SHA1: f50890ec55a0a4e23a89c721d0f90794ac300c66

- Once downloads are complete, go to the download folder and launch the QuartusLiteSetup-18.1.0.625-windows.exe application.
- Accept the agreement, choose the install directory (default C:\intelFPGA_lite\18.1 works fine), verify the correct components are being installed (see below) and begin the install.



- At the end of the install, choose the option to Launch USB Blaster II driver installation. This will install the USB driver to connect to the FPGA. Complete driver installation.

- a. If there is trouble installing USB Blaster, follow instructions here for your Windows version: https://www.terasic.com.tw/wiki/Altera_USB_Blaster_Driver_Installation_Instructions. The default driver path is C:\intelFPGA_lite\18.1\quartus\drivers



6. Launch Quartus (Quartus Prime 18.1) Lite Edition from the desktop. Once Quartus Prime Lite Edition loads, select Tools / Options from the toolbar. This will open the Options window.
7. In the Options window, select EDA Tool Options on the left-hand side. Click the "... " button to the right of the ModelSim-Altera directory field
8. In the Select Directory window, navigate to the directory where ModelSim has been installed (C:\intelFPGA\18.1\modelsim_ase) and choose the directory "win32aloem". Click Select Folder, then OK to close the Options window. Quartus Prime 18 and ModelSim are now installed.

